

SAGE: Source-Aware Multiframe Thermal Detection for Coastal Surveillance

Technical Report — controlled source-aware evaluation

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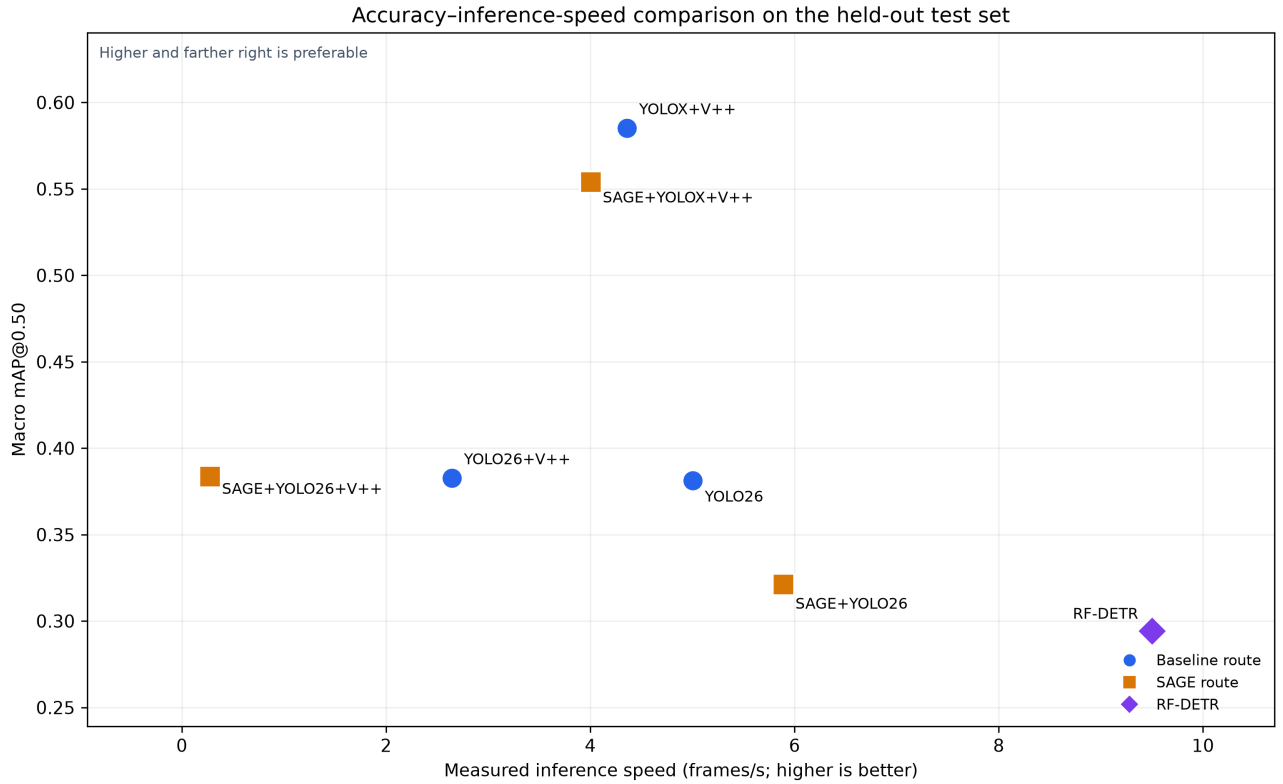


Figure 1: Accuracy–inference-speed comparison across all seven models. Each labelled marker reports macro mAP@0.50 and measured per-target throughput; colour and marker shape distinguish baseline detectors, SAGE detectors, and RF-DETR.

Abstract

Coastal thermal surveillance must detect small vessels, people, swimmers, and buoys when targets are faint, tiny, or confused with waves, reflections, shore-side structure, and sensor artefacts. A single generic preparation path is not well matched to both wide-field panoramic camera (PANO) footage and forward-looking infrared (FLIR) footage with changing polarity, scale, and scene structure. This paper evaluates Source-Aware Gated Enhancement (SAGE), which detectors each frame

using recorded PANO/FLIR metadata before a shared causal multiframe detector. For PANO, the Adaptive Global-Context Tensor (AGCT) strengthens useful local contrast while retaining the surrounding scene; for FLIR, the Weighted Scene-Guided Tensor (WSGT) standardises the image presentation and weighs target-like detail against broader scene structure.

SAGE produces a consistent three-part representation for every frame and supplies five chronological frames to direct YOLO26-P2 or YOLOV++ temporal detectors. The study compares three matched baseline/SAGE pairs on a frozen 542-clip held-out test set, with RF-DETR as an open-source comparator. The contribution is a controlled source-aware architecture and evaluation, not a claim of universal accuracy improvement. Direct SAGE plus YOLO26-P2 improves Person, whereas the temporal SAGE detectors are neutral. The inference speed of the YOLOX-YOLOV++ detector has the highest inference speed paired with mAP50 followed by its SAGE version.

1 Introduction

1.1 Problem and importance

This paper addresses source-dependent failure modes in coastal thermal detection. PANO video can make fast vessels small in a wide field of view and visible in only a few sampled frames. FLIR video can change target and clutter appearance through polarity, zoom, region of interest, and stable scene structure. The open operational problem is to detect the same deployment classes across these conditions without redesigning the downstream detector for each camera source.

1.2 Why existing methods remain insufficient

Hand-designed contrast, segmentation, and candidate-region methods can highlight dim contacts or suppress known clutter, but fixed thresholds and scales are sensitive to waves, reflections, polarity, zoom, and unfamiliar backgrounds [6]. Learned single-frame detectors improve multi-scale representation [1, 2], yet remain dependent on the source, target scale, and clutter distribution represented in training. Temporal aggregation can recover weak evidence [3], while cross-modal and scene-aware methods add useful representation or context but often require aligned modalities, scene masks, or more complex deployment paths [5, 6].

1.3 Goal, approach, constraints, and contributions

The goal is to test whether source-aware preparation provides useful information beyond matched spatial and temporal detectors. SAGE uses recorded source metadata to select a deterministic PANO or FLIR branch, but does not learn the camera tag as a detector feature, use future frames, or change the detector output labels. Both branches create the same ordered three-part input, allowing one causal five-frame detector interface.

The approach therefore combines deterministic PANO/FLIR routing to address the representation mismatch caused by a single generic preparation path with a causal multiframe YOLO26-P2 implementation that receives five chronological frames as a 15-channel input while predicting only the newest frame. It uses matched data, labels, thresholds, and held-out clips to separate preparation effects from protocol changes, and keeps detector mAP separate from the unproven coverage hypothesis for source routing and a single-frame recovery path.

2 Previous Work

Thermal maritime object detection has been addressed with both hand-crafted and learned approaches [6]. Earlier infrared methods use sea/land or cloud segmentation, local contrast, centre-surround filtering, sparse or low-rank decomposition, and hand-crafted texture or shape features to expose a small target and suppress background responses. More recent data-driven detectors learn these features from labelled examples: one-stage YOLO detectors emphasise low-latency box localisation, whereas two-stage and transformer detectors invest more computation in refinement or global relationships. Feature pyramids and high-resolution heads retain fine spatial detail alongside semantic features, helping small vessels, people, and buoys survive downsampling; YOLO26 [1] and YOLOX [2] are practical real-time detector families in this line of work.

Other work improves the representation supplied to the detector or adds context that is unavailable in one still image. Pseudo-visible and visible-thermal transfer methods, including PFGF [5], seek complementary cross-modal information when thermal texture is weak, while scene-aware multi-task methods such as SMPISD-MTPNet [6] use scene semantics, multi-scale fusion, enhancement, and auxiliary segmentation to reduce structured-background false alarms. Video detection adds a further source of evidence: YOLOV++ [3] selects and aggregates features from nearby frames so that persistent, blurred, occluded, or intermittently visible contacts can support the target-time prediction.

Across these approaches, the same limitations remain. Low-contrast contacts, wakes, waves, clouds, shore-side structures, and reflections can share target-like cues; changing target scale, polarity, zoom, field of view, and background texture can defeat fixed filters or shift learned distributions [6]. Learned methods depend on representative target and clutter examples, while scene-aware and cross-modal detectors can require extra masks or aligned modalities [5, 6]. Temporal components introduce additional feature-selection and aggregation operations [3]. These constraints motivate source-stratified evaluation rather than assuming that success on a benchmark or a narrow scene establishes robust performance across sensors, viewpoints, and conditions.

3 Method

The study evaluates three video-based baseline detectors and then pairs each one with SAGE: (1) direct YOLO26-P2 detection, (2) YOLO26-P2 with YOLOV++ temporal aggregation, and (3) YOLOX with YOLOV++ temporal aggregation. The baselines share the same chronological input contract and newest-frame prediction rule; SAGE changes only source-aware frame preparation before the applicable baseline route.

3.1 Method overview and architecture

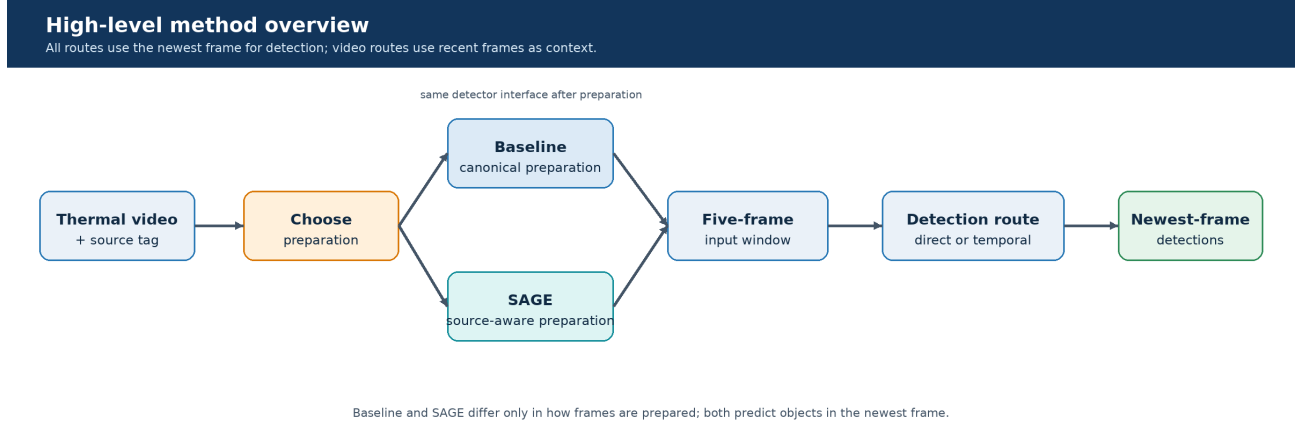


Figure 2: Study architecture overview.

Figure 2 presents the shared high-level flow. A chronological thermal-video input and its recorded source tag first determine whether frames receive canonical preparation or SAGE source-aware preparation. Both choices produce the same five-frame input window, which then enters a direct or temporal detection route. Every route returns detections for the newest frame only. The next three sections describe the baseline detectors separately, followed by the shared SAGE preparation layer.

3.2 Baseline A: Direct YOLO26-P2 detection

The first baseline is direct YOLO26-P2 detection. YOLO26-P2 is a one-stage detector with an additional high-resolution P2 detection level [1]. The P2 head operates at stride 4, alongside P3, P4, and P5 at strides 8, 16, and 32, retaining fine detail for distant or small targets.

For this route, each sample is a causal five-frame window ($t-4$, $t-3$, $t-2$, $t-1$, t). The frame-packing adapter converts each prepared frame into three ordered channels and concatenates the five frames into one 15-channel input. The first convolution is expanded accordingly, while the backbone, neck, P2–P5 heads, and target-time objective remain the YOLO26-P2 design. Earlier frames supply context; only the newest frame t is predicted.

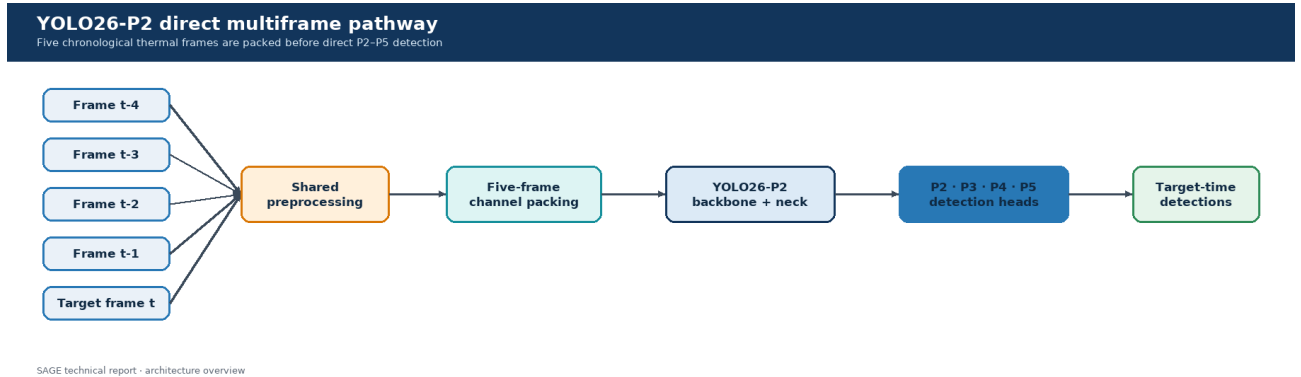


Figure 3: Direct YOLO26-P2 multiframe baseline.

3.3 Baseline B: YOLO26-P2 with YOLOV++ temporal aggregation

The second baseline retains the YOLO26-P2 spatial detector and adds YOLOV++ temporal aggregation after spatial feature extraction [3]. Chronological batches are constructed from consecutive packed samples. The YOLO26 spatial detector, backbone, neck, P2–P5 feature pyramid, and detection head are preserved; the temporal adapter is inserted between the feature pyramid and the existing detection head.

The temporal adapter selects and combines useful P2–P5 features from recent samples, then provides enhanced target-time features to the unchanged detection head. This route therefore tests whether feature-level temporal aggregation adds value beyond direct five-frame input fusion without changing the core YOLO26-P2 detector. It remains causal: frames after t are never used.

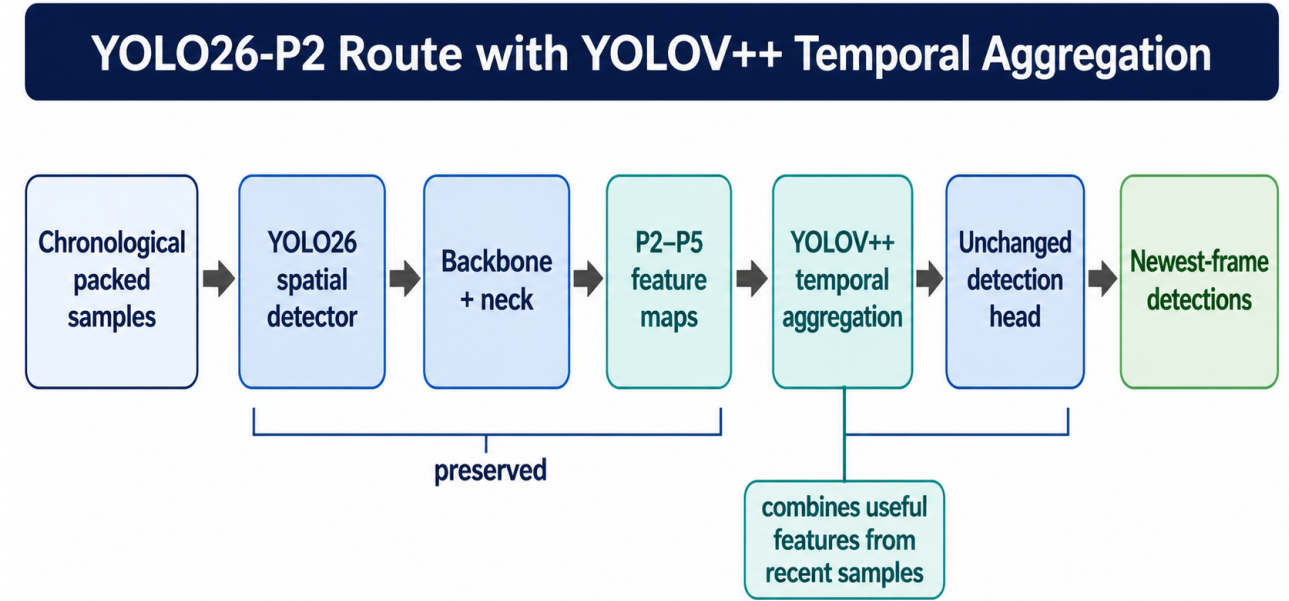


Figure 4: YOLO26-P2 with YOLOV++ temporal aggregation baseline.

3.4 Baseline C: YOLOX with YOLOV++ temporal aggregation

The third baseline begins with YOLOX [2] trained on the single-frame dataset, where each training example is one labelled target image. This stage learns the spatial detector weights from individual-frame appearance, scale, and background evidence. Those YOLOX weights are then transferred to the YOLOV++ implementation [3], which is trained on the multiframe dataset: chronological packed samples ending at the labelled newest frame t .

This staged design gives the temporal route a spatial detector that already recognises single-frame target patterns before YOLOV++ learns how recent P2–P5 features should be selected and combined. It separates spatial initialization from temporal learning, rather than asking the multiframe model to learn both from scratch. During multiframe training and inference, only frames up to t are used; YOLOV++ aggregates relevant recent features before the detection head returns the newest-frame

prediction. The YOLOX and YOLOV++ repositories cited in [2, 3] provide the respective detector and temporal-aggregation implementations.

This route provides an architecture-diverse temporal baseline. It keeps the same causal timing, post-processing, and held-out protocol as the other detectors, while changing the spatial-detector family. Comparing it with its SAGE-paired condition therefore tests whether source-aware preparation behaves consistently when the spatial representation is different.

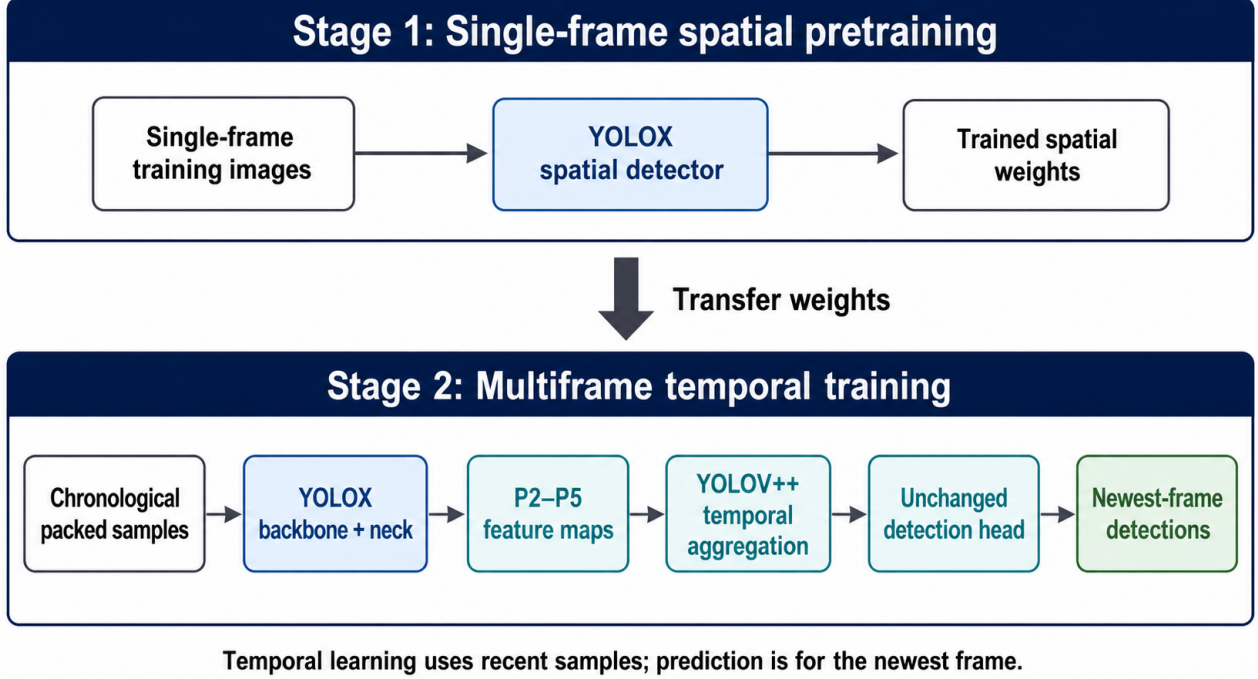


Figure 5: YOLOX-to-YOLOV++ staged training baseline.

3.5 SAGE source-aware preparation shared by all paired conditions

SAGE is a deterministic, source-routed preparation layer paired with each baseline. For a thermal frame $X_t \in [0, 1]^{H \times W}$ and recorded source tag $s_t \in \{\text{PANO}, \text{FLIR}\}$, SAGE applies

$$\mathbf{R}_t = \begin{cases} \mathbf{R}_{\text{PANO}}(X_t), & s_t = \text{PANO}, \\ \mathbf{R}_{\text{FLIR}}(X_t), & s_t = \text{FLIR}. \end{cases}$$

The source tag selects the preparation branch only. It is not supplied to the detector as a learned feature. Both branches return the same ordered three-channel representation, $\mathbf{R}_t = [X_t, \tilde{X}_t, M_t] \in [0, 1]^{3 \times H \times W}$, where X_t is the original-looking intensity view, $\tilde{X}_t$ is an enhanced or polarity-normalised view, and M_t is a source-specific cue map. This fixed channel meaning allows all the downstream models to use the same input interface.

3.5.1 PANO branch: Adaptive Gated Contrast Tensor (AGCT)

For PANO frames, AGCT applies tile-local contrast enhancement only when whole-frame statistics indicate that it is useful. Let $X_t^{(g)}$ denote a grayscale copy of the frame. Robust contrast and entropy

are $q_t = P_{95}(X_t^{(g)}) - P_{05}(X_t^{(g)})$ and $H_t = -\sum_{i=0}^{255} p_i \log_2 p_i$, where p_i is the empirical 256-bin intensity histogram. The contrast and entropy gates are

$$g_q = \sigma\left(k_q \frac{q_{\text{hi}} - q_t}{q_{\text{hi}} - q_{\text{lo}}}\right), \quad g_H = \sigma\left(k_H \frac{H_{\text{hi}} - H_t}{H_{\text{hi}} - H_{\text{lo}}}\right), \quad \alpha_t = g_q g_H,$$

where $\sigma(z) = (1 + e^{-z})^{-1}$. The values $q_{\text{lo}}, q_{\text{hi}}, H_{\text{lo}}, H_{\text{hi}}, k_q$, and k_H are selected from PANO training data and fixed before validation and held-out testing. Let $\mathcal{E}(\cdot)$ be the calibrated tile-local contrast transform. The enhanced PANO channel and cue map are

$$\tilde{X}_t^{\text{PANO}} = (1 - \alpha_t)X_t + \alpha_t \mathcal{E}(X_t), \quad M_t^{\text{PANO}} = \alpha_t \mathbf{1}_{H \times W},$$

so $\mathbf{R}_{\text{PANO}}(X_t) = [X_t, \tilde{X}_t^{\text{PANO}}, M_t^{\text{PANO}}]$. AGCT is therefore an adaptive blend rather than a fixed enhancement: it can expose faint PANO structure while retaining the original presentation when enhancement is not supported by the frame’s global statistics.

3.5.2 FLIR branch: Whole-scene and Scale-Gated Glint Tensor (WSGT)

FLIR footage was assessed under both *white-hot* and *black-hot* display polarity. A target or clutter feature may therefore appear either brighter or darker than its local background. WSGT uses a polarity-invariant local-deviation response so that its cue is responsive to both black-hot and white-hot displays. For a local neighbourhood $\mathcal{N}(u)$ around pixel u , let $\mu(u)$ and $\sigma(u)$ be the local mean and standard deviation. The local standardised magnitude and polarity response are

$$z_t(u) = \frac{|X_t(u) - \mu(u)|}{\sigma(u) + \varepsilon}, \quad r_t(u) = \sigma(k_z[z_t(u) - \tau_z]),$$

where k_z is a fixed gate slope and τ_z is the training-calibrated local-deviation threshold. The absolute-value term ensures equivalent treatment of locally bright responses in white-hot imagery and locally dark responses in black-hot imagery. WSGT combines the polarity response with whole-scene compatibility and differentiable scale guidance:

$$M_t^{\text{FLIR}}(u) = r_t(u) c_\psi(X_t) \omega_s(a_t(u)), \quad \omega_s(a) = \sigma(\gamma(a - a_{\min})) \sigma(\gamma(a_{\max} - a)).$$

Here $c_\psi(X_t) \in [0, 1]$ is a learned whole-frame environmental-compatibility prior, and ω_s is a differentiable two-sided gate applied to local scale proxy $a_t(u)$. The scale bounds $a_{\min}$ and $a_{\max}$, along with slope γ , are selected from training annotations under the project image-resize policy and fixed before validation and test inference. Candidate-component checks remain offline diagnostics; they are not represented as non-differentiable operations in the training tensor. The FLIR representation is $\mathbf{R}_{\text{FLIR}}(X_t) = [X_t, \tilde{X}_t^{\text{FLIR}}, M_t^{\text{FLIR}}]$, where $\tilde{X}_t^{\text{FLIR}}$ is the polarity-normalised intensity view. Temporal aggregation is not performed within WSGT. YOLOV++, where used, operates later on P2–P5 detector features.

3.5.3 Fixed interface and causal multiframe input

For each target time t , SAGE concatenates the five chronologically prepared frames:

$$\mathbf{Z}_t = \text{concat}(\mathbf{R}_{t-4}, \mathbf{R}_{t-3}, \mathbf{R}_{t-2}, \mathbf{R}_{t-1}, \mathbf{R}_t) \in \mathbb{R}^{15 \times H \times W}.$$

All SAGE conditions therefore retain the same causal, $t_4 - t$, 15-channel interface as their matched baselines and predict only frame t . SAGE changes the frame representation only, it does not change the detector labels, spatial detector, temporal module, split manifest, post-processing, or threshold-selection procedure.

3.5.4 Contribution

SAGE contributes a fixed-interface source-routing design: AGCT uses global frame statistics to control PANO contrast enhancement, while WSGT accounts for white-hot and black-hot FLIR polarity through a polarity-invariant local response, then combines whole-scene and scale-aware guidance. Both branches produce the same causal, $t_4 - t$, three-channel representation for all the detectors.

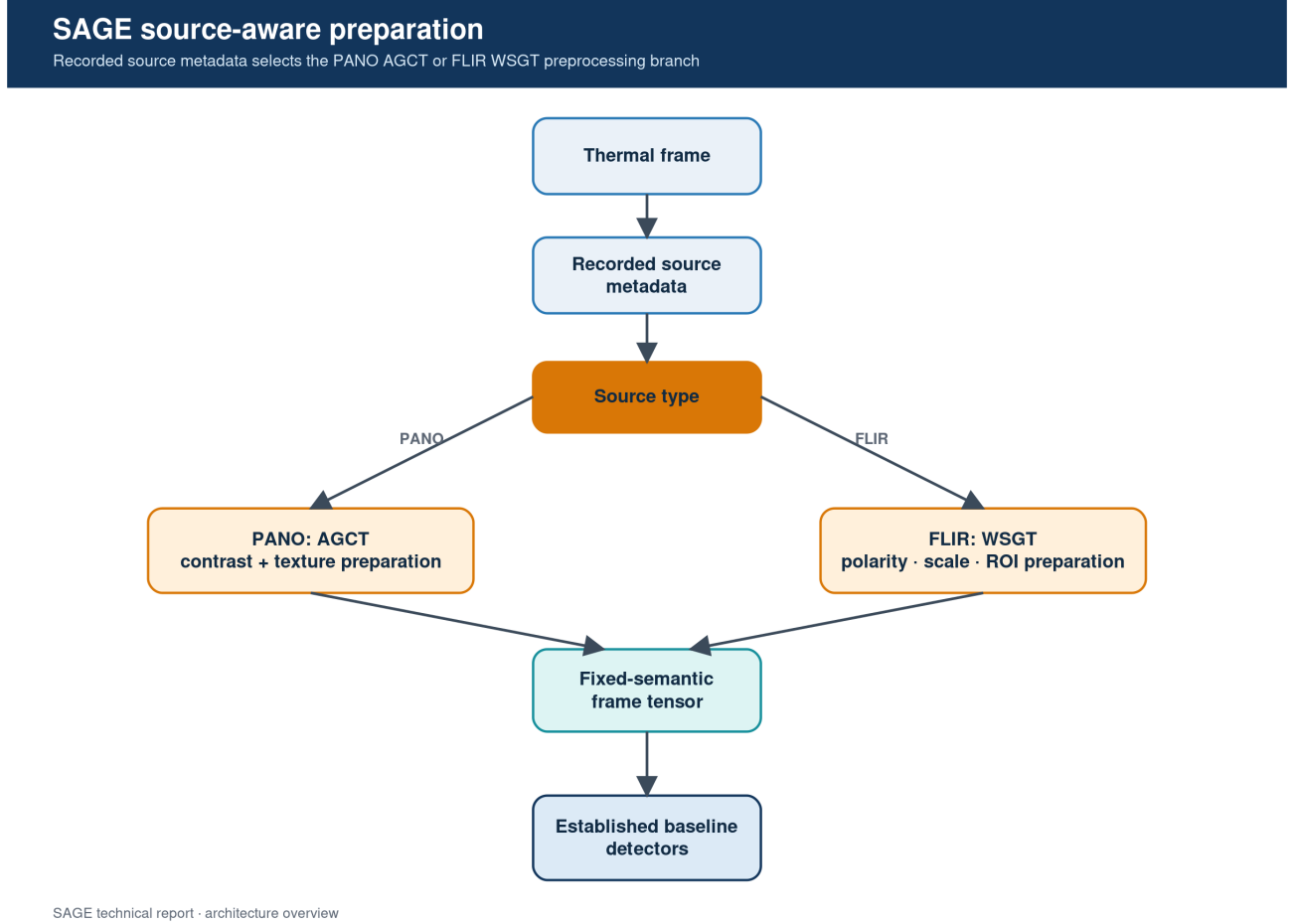


Figure 6: SAGE source-aware preparation shared by all paired conditions.

3.6 Output and implementation boundary

All detectors output boxes, confidence scores, and superclass scores for frame t . The same non-maximum suppression and validation-selected confidence threshold are applied to each baseline/SAGE pair. The evaluation is therefore a causal video-detection comparison, not an image-enhancement-only evaluation or an unrestricted video model.

3.7 Algorithm summary

For every baseline sample, construct a causal window from $t-4$ through t . Baseline conditions use canonical preparation, while SAGE conditions select AGCT or WSGT from the recorded source tag.

Baseline A sends the packed 15-channel tensor directly to YOLO26-P2. Baseline B preserves YOLO26-P2 and applies YOLOV++ to its P2–P5 features. Baseline C applies the same temporal principle with YOLOX as the spatial detector. All detectors return detections for t , with validation-fixed thresholds used for held-out testing.

4 Experimental Results

4.1 Dataset, protocol, and implementation details

4.1.1 Single-frame and multiframe inputs

The image-based RF-DETR comparator [4] receives one thermal image at a time and returns detections for that same image. It provides a still-image reference without using information from earlier or later images.

The video-based detectors receive a short chronological sequence ending at the current image. Earlier images provide context for the current prediction, while detections are returned only for the newest image. This makes the comparison easy to interpret: image-based detection relies on one view, whereas video-based detection can use recent motion and persistence.

4.1.2 Training and inference consistency

All baseline/SAGE pairs must use the same raw sequences, packed-frame protocol, split manifest, image-size policy, training schedule, validation selection rule, threshold-selection procedure, and held-out evaluation. Geometric and intensity augmentation must be shared across every frame within a packed sample. Future frames must not be used for a target-time prediction.

For the documented YOLO26-P2 route with YOLOV++ temporal aggregation, which combines the YOLO26 spatial detector [1] with YOLOV++ temporal aggregation [3], chronological batches are constructed from consecutive packed samples. The temporal adapter preserves the spatial backbone, neck, and detection head while aggregating P2–P5 features. Any implementation difference between the YOLOX and YOLO26-P2 detectors with YOLOV++ temporal aggregation must be recorded in the run manifest.

4.1.3 Matched comparison design

Each SAGE condition is matched only to its corresponding baseline:

Pair	Baseline	SAGE condition	Primary question
P1	YOLO26-P2	SAGE + YOLO26-P2	Does source-aware preparation add value to direct multiframe detection?
P2	YOLO26-P2 with YOLOV++ temporal aggregation	+ SAGE YOLOV++ temporal aggregation	Does SAGE add value after temporal feature aggregation?
P3	YOLOX with YOLOV++ temporal aggregation	+ SAGE YOLOV++ temporal aggregation	Does the effect persist with a different spatial initialization path?

Table 1: Matched baseline/SAGE comparison design.

4.1.4 Evaluation protocol

Validation selects checkpoints and operating thresholds; the held-out test is used only after configuration freeze. Report precision, recall, F1, mAP@0.50, mAP@0.50:0.95, class-wise AP/recall, false positives per minute, target-frame latency, throughput, peak memory, and SAGE overhead. Bootstrap complete source sequences—not individual correlated frames—for confidence intervals.

4.2 Baselines and comparators

ID	Model	Spatial initialization / detector	Temporal mode	SAGE	Held-out status
B1	YOLO26-P2	YOLO26-P2	Direct five-frame early fusion	No	Completed
B2	YOLO26-P2 with YOLOV++ temporal aggregation	YOLO26-P2	YOLOV++ temporal aggregation	No	Completed
B3	YOLOX with YOLOV++ temporal aggregation	YOLOX	YOLOV++ temporal aggregation	No	Completed
S1	SAGE + YOLO26-P2	YOLO26-P2	Direct five-frame early fusion	Yes	Completed with PANO/FLIR routing
S2	SAGE + YOLO26-P2 with YOLOV++ temporal aggregation	YOLO26-P2	YOLOV++ temporal aggregation	Yes	Completed with PANO/FLIR routing
S3	SAGE + YOLOX with YOLOV++ temporal aggregation	YOLOX	YOLOV++ temporal aggregation	Yes	Completed with PANO/FLIR routing
R1	RF-DETR	RF-DETR	Single-frame open-source comparator	No	Completed; reported as four superclasses

Table 2: Baseline and comparator configurations.

The six B/S rows form the controlled internal study. RF-DETR [4] is the image-based open-source comparator: it uses a read-only COCO adapter for individual images and annotations, without a temporal window, 15-channel packer, SAGE, or YOLOV++ adapter. For shared reporting, its validation and test inference use the same target-frame manifests as the multiframe detectors, and results are reported as the four approved superclasses.

4.3 Quantitative results

All held-out values use the frozen held-out test manifest. Each model’s confidence threshold was selected on validation only and then held fixed for test inference. This report discloses only Boat, Person, Swimmer, and Buoy. Boat and Person AP are instance-weighted aggregates.

For the size benchmark, object area is measured in original-image pixels using the COCO area ranges [7]: Small ($A < 32^2$ px²), Medium ($32^2 \leq A < 96^2$ px²), and Large ($A \geq 96^2$ px²). All predictions use each model’s validation-selected test threshold. For a selected area band, a same-class prediction that overlaps an object outside that band at $\text{IoU} \geq 0.50$ is ignored rather than counted as a false positive. Superclass mAP@0.50 is the instance-weighted mean of its member-class APs. Speed is CUDA-synchronised model inference plus the report’s superclass NMS on an NVIDIA RTX A6000, excluding image loading and preprocessing.

4.3.1 All models on the approved superclasses

Type	Model	Boat@50	Boat@50:95	Person@50	Person@50:95	Swimmer@50	Swimmer@50:95	Buoy@50	Buoy@50:95
Image-based	RF-DETR	0.630	0.412	0.565	0.289	0.511	0.182	0.133	0.042
Video-based	YOLO26-P2	0.603	0.373	0.319	0.187	0.693	0.345	0.295	0.182
Video-based	YOLO26-P2 + YOLOV++	0.574	0.359	0.396	0.214	0.678	0.359	0.247	0.156
Video-based	YOLOX + YOLOV++	0.884	0.616	0.866	0.590	0.959	0.505	0.537	0.283
Video-based	SAGE + YOLO26-P2	0.528	0.305	0.450	0.210	0.500	0.188	0.145	0.080
Video-based	SAGE + YOLO26-P2 + YOLOV++	0.572	0.358	0.388	0.208	0.677	0.360	0.257	0.165
Video-based	SAGE + YOLOX + YOLOV++	0.835	0.564	0.764	0.493	0.576	0.243	0.518	0.275

Table 3: Table 1. Held-out superclass mAP@0.50 and mAP@0.50:0.95 for all compared models.

RF-DETR uses a global confidence threshold of 0.206 selected on validation and held fixed for the held-out test.

4.3.2 Size-stratified superclass mAP50 benchmark

Model	Boat S	Boat M	Boat L	Person S	Person M	Person L	Swimmer S	Swimmer M	Swimmer L	Buoy S	Buoy M	Buoy L	FPS
RF-DETR	0.059	0.519	0.779	0.454	0.479	0.263	0.176	0.524	0.105	0.049	0.124	0.000	9.502
YOLO26M-P2	0.436	0.644	0.348	0.103	0.515	0.223	0.875	0.674	0.000	0.755	0.000	0.000	5.002
YOLO26M-P2 + YOLOV++	0.358	0.625	0.324	0.116	0.584	0.163	0.953	0.724	0.000	0.745	0.000	0.000	2.644
YOLOX + YOLOV++	0.281	0.648	0.771	0.431	0.653	0.444	0.944	0.948	0.990	0.545	0.223	0.143	4.359
SAGE + YOLO26M-P2	0.177	0.570	0.497	0.390	0.468	0.218	0.502	0.578	0.000	0.455	0.000	0.000	5.890
SAGE + YOLO26M-P2 + YOLOV++	0.377	0.629	0.330	0.109	0.581	0.200	0.912	0.717	0.000	0.748	0.000	0.000	0.274
SAGE + YOLOX + YOLOV++	0.256	0.636	0.758	0.486	0.664	0.413	0.950	0.883	0.825	0.550	0.229	0.000	4.003

Table 4: Table 2. Held-out, size-stratified superclass mAP@0.50 and calibrated per-target throughput. S/M/L denote COCO small/medium/large area bands; speed is frames per second.

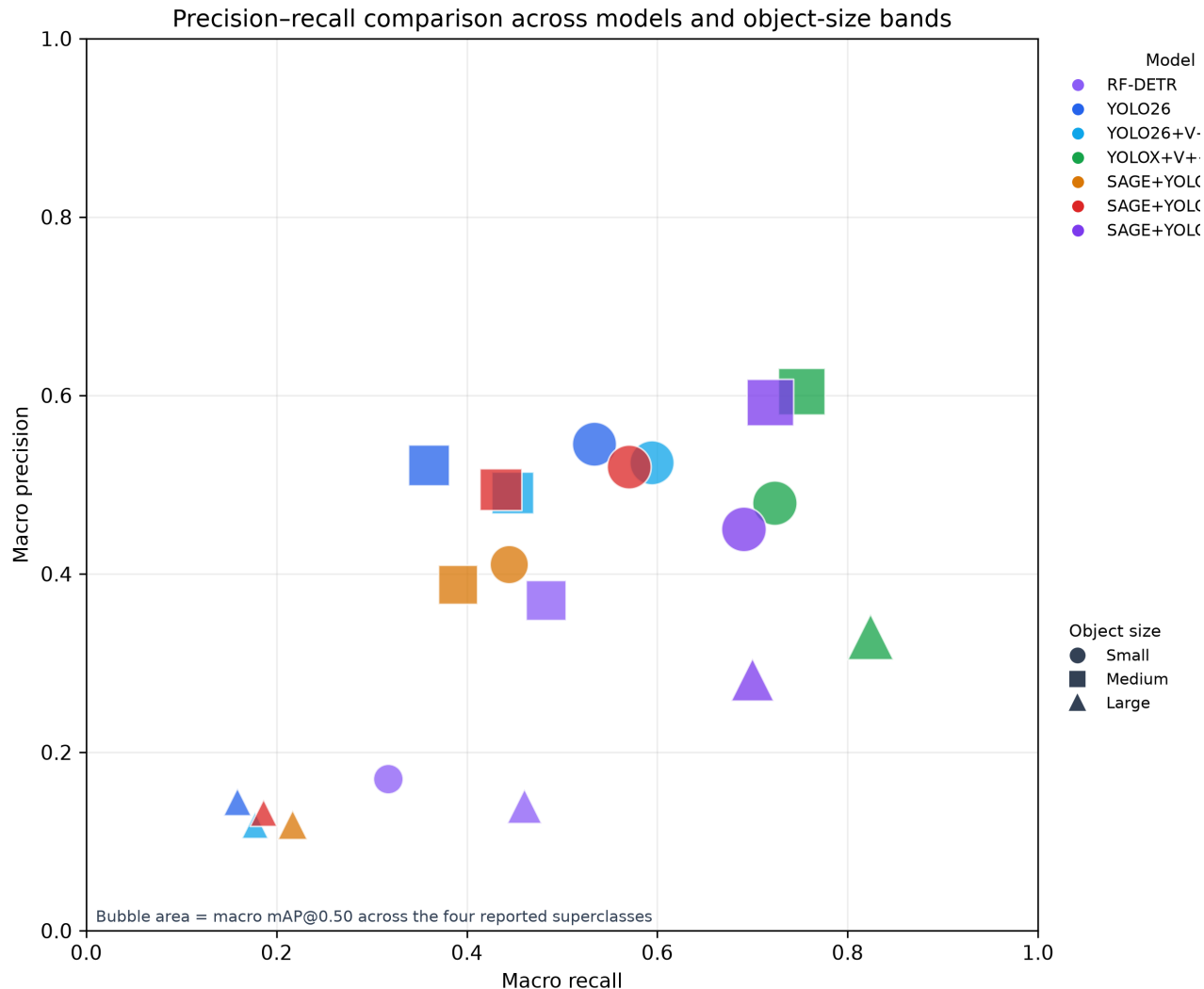


Figure 7: Precision-recall operating points for all seven models and Small/Medium/Large object-size bands in one chart. Colour identifies the model, marker identifies the size band, and bubble area is macro mAP@0.50 across the four reported superclasses.

4.4 Quantitative discussion

The central result is architecture-dependent rather than uniformly source-preparation-dependent. YOLOX with YOLOV++ is strongest on all four reported superclasses. Direct SAGE plus YOLO26-P2 improves Person mAP50 by 0.131 but decreases Boat, Swimmer, and Buoy. The matched YOLO26-P2 temporal pair is practically unchanged, while SAGE plus YOLOX with YOLOV++ is lower on every reported superclass, including a 0.384 decrease for Swimmer. RF-DETR is strongest among the non-YOLOX detectors on Person mAP50, but it is not the leading model overall and is a single-frame open-source comparator rather than a temporal replacement.

4.5 Qualitative evaluation and ablation status

This version includes architecture diagrams and quantitative held-out results, but not a systematic visual-error gallery or AGCT-only/WSGT-only ablation table. Those analyses are required before attributing an outcome to an individual branch. Future work should show matched true positives, misses, and false positives by source and error type, then hold route, split, threshold rule, and post-processing fixed while comparing canonical preparation, AGCT-only, WSGT-only, and full SAGE. Complete-event trials should also measure missed contacts, alert timing, route availability, and recovery-path contribution.

5 Conclusion

This paper presented SAGE, a source-aware preparation method for causal five-frame thermal detection. It routes PANO and FLIR frames through deterministic branches, converts both into a common three-part representation, and preserves one 15-channel detector interface for direct YOLO26-P2 and YOLOV++ detectors. Its contributions are the controlled input design, implementation-grounded multiframe architecture, and matched held-out evaluation. Operationally, the same design supports deterministic PANO/FLIR routing while a multiframe primary route and an independent single-frame recovery route can run in parallel, preserving a recovery opportunity when sparse PANO sampling leaves a fast vessel visible in only a few frames.

The results do not support SAGE as a universal accuracy improvement. YOLOX with YOLOV++ is strongest across Boat, Person, Swimmer, and Buoy while direct SAGE plus YOLO26-P2 improves Person but reduces the other superclasses and the YOLO26-P2 temporal pair is effectively neutral. Its evidence is instead consistent with a targeted source-aware representation that can improve selected small-target regimes, especially Person, Swimmer and Buoy detection, at a modest inference-speed cost. Therefore, source-stratified branch ablations and complete-event trials are required to evaluate operational coverage.

Acknowledgements

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References

- [1] G. Jocher, J. Qiu, M. Liu, S. Lyu, F. C. Akyon, and M. E. Kalfaoglu, “Ultralytics YOLO26: Unified real-time end-to-end vision models,” 2026, arXiv:2606.03748; doi: 10.48550/arXiv.2606.03748. Official implementation: <https://github.com/ultralytics/ultralytics>. [Online]. Available: <https://arxiv.org/abs/2606.03748>
- [2] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, “YOLOX: Exceeding YOLO series in 2021,” 2021, arXiv:2107.08430; doi: 10.48550/arXiv.2107.08430. Official implementation: <https://github.com/Megvii-BaseDetection/YOLOX>. [Online]. Available: <https://arxiv.org/abs/2107.08430>

- [3] Y. Shi, T. Zhang, and X. Guo, “Practical video object detection via feature selection and aggregation,” 2024, arXiv:2407.19650; YOLOV++; official implementation: <https://github.com/YuHengsss/YOLOV>. [Online]. Available: <https://arxiv.org/abs/2407.19650>
- [4] I. Robinson, P. Robicheaux, M. Popov, D. Ramanan, and N. Peri, “RF-DETR: Real-time detection transformer,” in *International Conference on Learning Representations*, 2026, official implementation: <https://github.com/roboflow/rf-detr>. [Online]. Available: <https://arxiv.org/abs/2511.09554>
- [5] T. Li, M. Ye, T. Wu, N. Li, S. Li, S. Tang, and L. Ji, “Pseudo visible feature fine-grained fusion for thermal object detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 6710–6719, doi: 10.1109/CVPR52734.2025.00629. Official implementation: <https://github.com/liting1018/PFGF>. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025/html/Li_Pseudo_Visible_Feature_Fine-Grained_Fusion_for_Thermal_Object_Detection_CVPR_2025_paper.html
- [6] C. Hu, X. Dong, Y. Huang, L. Wang, L. Xu, T. Pu, and Z. Peng, “SMPISD-MTPNet: Scene semantic prior-assisted infrared ship detection using multitask perception networks,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–14, 2025, doi: 10.1109/TGRS.2024.3516879. Official implementation: <https://github.com/greekinRoma/SMPISD-MTPNet>. [Online]. Available: <https://arxiv.org/abs/2407.18487>
- [7] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, “Microsoft COCO: Common objects in context,” in *Computer Vision – ECCV 2014*. Springer International Publishing, 2014, pp. 740–755, doi: 10.1007/978-3-319-10602-1_48. [Online]. Available: https://doi.org/10.1007/978-3-319-10602-1_48